

CASE STUDY

AI in Seasonal Decor Manufacturing and Distribution

Supply Chain Intelligence

introduced warehouse orchestration views that linked forecasted demand to inbound congestion, wave planning, cut-off risk, and labor visibility.

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Problem

The client's seasonal décor manufacturing and distribution business was under pressure from highly compressed demand windows, long and variable supplier lead times, and rapid post-season inventory value loss. Unlike steady-state retail, the business had to buy, build, and position the right SKUs months in advance, then execute perfectly during a short holiday peak. A small forecasting error could mean missing the selling season entirely, while overbuying could leave the team with aged stock, markdown exposure, and cash tied up in inventory that would not sell profitably after the peak.

Operational data was also fragmented across sales, procurement, warehouse, and supplier systems. The team had limited visibility into which products were most at risk, which suppliers were slipping, how much safety stock was actually needed, and where warehouse congestion could block fulfillment. As a result, decisions were often reactive: rush buys, late escalations, excess stock, and missed shipment cutoffs during the most important sales periods of the year.

How we solved it

Innovacio Technologies designed an AI-powered seasonal control tower that unified sales signals, purchase orders, supplier performance, inventory aging, and warehouse execution into one dashboard. This gave merchandising, procurement, inventory, and operations teams a single view of peak-season readiness, stock exposure, and fulfillment risk across the business.

We implemented SKU- and channel-level demand forecasting so the team could translate prior-season sales, calendar effects, promo plans, and sell-through patterns into actionable buy signals and replenishment targets. The dashboard also surfaced forecast confidence, bias, and exception alerts so planners could adjust early instead of waiting until the season was already underway.

To protect supply continuity, we added supplier risk scoring, lead-time prediction, and escalation logic so buyers could identify delayed orders sooner and shift to alternate sources when needed. We also built inventory aging and safety-stock controls that distinguished between pre-season build quantities, in-season protection stock, and post-season liquidation risk.

For execution, we introduced warehouse orchestration views that linked forecasted demand to inbound congestion, wave planning, cut-off risk, and labor visibility. The result was a more coordinated seasonal operating model where forecasting, procurement, inventory, and fulfillment



moved together, helping the client reduce stockouts, avoid excess aged inventory, and execute peak season with greater confidence.

Executive summary

Seasonal décor businesses live and die by the calendar. Demand rises sharply ahead of major holiday windows, then falls quickly after the peak. That makes procurement timing, inventory depth, warehouse slotting, and markdown planning unusually hard to manage with spreadsheets alone.

AI is valuable here because the main problem is not just forecasting demand. It is turning a seasonal forecast into the right purchase orders, the right stock mix, the right safety stock, and the right fulfillment plan across factories, suppliers, and distribution centers.

This case study shows how Innovacio would solve that problem with a role-based AI supply-chain dashboard that connects sales signals, purchase orders, supplier risk, inventory aging, and warehouse execution into one operating view.

Why this industry is different

Demand is highly seasonal and calendar-driven, so a model that works in March may fail badly in Q4.

Lead times can be long and variable because products may rely on imported components, packaging, bulbs, electronics, molded parts, or freight-constrained shipments.

Inventory risk is asymmetric: buying too little causes stockouts during the peak window, while buying too much creates heavy post-season markdowns and obsolete stock.

Procurement must handle many SKUs, suppliers, and product variants, which makes manual reordering fragile and slow.

Research basis

The research consistently points to four themes: seasonal demand forecasting, inventory control that explicitly accounts for seasonality, near-term stockout prediction using real-time inventory data, and tighter integration between forecasting and replenishment decisions. Studies on seasonal retail demand and inventory control show that ignoring seasonality creates avoidable

optimality gaps, while recent AI/ML work in supply chains emphasizes supplier selection, inventory control, demand estimation, and replenishment planning as high-value use cases.

Reusable AI map for seasonal décor

Area	What the team needs	What AI adds	Operational effect
Demand	Calendar signals, promo plans, historical sales, weather, events	Forecasting by SKU/channel/week	Better peak planning
Procurement	Supplier lead times, MOQ, freight, PO history	Risk scoring and reorder suggestions	Fewer late buys
Inventory	Stock on hand, aging, locations, reserve stock	Safety stock logic and alerting	Lower excess and stockouts
Fulfillment	DC capacity, pick rates, order cutoffs	Wave planning and exception alerts	Smoother peak execution

Predictive demand planning for holiday peaks

Business problem: The team faced sharp Q4 demand spikes across ornaments, décor bundles, lighting, outdoor products, and accessory SKUs. Forecast errors were expensive because the buying window for seasonal merchandise is narrow: if the plan is late, the peak season starts before inventory arrives; if the plan is too high, the company is forced into post-season discounting.

Data problem: Historical sales alone were not enough. The team needed to combine prior-year sell-through, calendar effects, event timing, promo calendars, product life cycles, channel mix, and shipment lead times. Research on seasonal retail forecasting shows that special days and holiday periods produce demand patterns that are materially different from ordinary days, so one flat forecast is not sufficient.

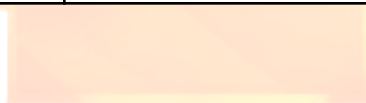


AI solution: Innovacio built a demand-forecasting layer that forecasts by SKU family, channel, and time bucket, then decomposes the result into buy signals, safety stock targets, and replenishment alerts. The dashboard exposes forecast confidence, bias, sell-through curves, and exception flags so merchandising and procurement teams can act before the season window opens.

Why it matters: Studies on seasonal forecasting and inventory control show that accounting for seasonality improves decision quality, while recent work on forecasting methods for high seasonal variation demonstrates that short-term forecasting models are most useful when the product demand pattern changes with the season. This is exactly the setting where a dashboard-driven AI layer is more valuable than manual spreadsheets.



Metric	Before	After AI planning	What changed
Forecast process	Manual spreadsheet planning	SKU/channel forecast dashboard	Faster decisions
Safety stock	Static rules	Seasonality-aware safety stock	Less stockout risk
Buy timing	Reactive reorders	Forward buy signals	Better arrival timing



Supplier risk and procurement orchestration

Business problem: Seasonal décor often depends on multiple upstream suppliers for packaging, molded components, lighting elements, printed materials, and imported goods. Lead times shift as freight conditions, factory schedules, and customs delays change. When procurement works from disconnected email threads, the team learns about delays too late to protect the season.

Data problem: Purchase orders, supplier OTIF performance, landed cost, expediting history, fill rates, and exception reasons usually live in different systems. AI research in supply chain management highlights supplier selection, planning, and replenishment as core use cases,



while seasonal lead-time research shows that inventory control should anticipate cyclical lead-time patterns rather than assume stable lead times.

AI solution: Innovacio solved this with supplier scorecards, lead-time prediction, and auto-escalation rules. The dashboard ranks suppliers by risk, lateness, cost volatility, and historical reliability. Buyers see which orders need to be accelerated, which can remain on plan, and which should be swapped to alternate suppliers or split shipments.

Why it matters: In a seasonal business, the procurement system is really a time-risk management system. Research on supply-chain inventory control and reorder-point optimization shows that seasonality and logistics reliability should directly shape replenishment decisions, which is exactly what the procurement dashboard is designed to do.

Procurement layer	Before	After AI layer	Outcome
Supplier visibility	Fragmented emails and calls	Supplier risk dashboard	Earlier escalation
Lead times	Assumed constant	Predicted by supplier/category	Better buy dates
Expedites	Ad hoc firefighting	Exception rules + alerts	Less chaos

Inventory aging, overstock, and markdown control

Business problem: Seasonal décor inventory loses value quickly after the peak window. Without early signal, teams overbuy in the name of service level and then carry aged stock into the wrong part of the year. That hurts cash flow, warehouse capacity, and margins.

Data problem: The company needed a combined view of on-hand stock, inbound stock, aging buckets, reserve stock, sell-through velocity, and location-based exposure. Research on inventory control with seasonality shows that the inventory policy itself should change with the season, while stockout prediction studies show that near-term inventory and recent sales are among the strongest predictors of short-term risk.



AI solution: Innovacio built an inventory command center that flags overstock, understock, and aging inventory by product family and DC. The system recommends pre-season build quantities, in-season reorder points, and post-season liquidation paths, so inventory decisions are tied to the actual demand curve rather than a static min-max rule.

Why it matters: Research on demand forecasting and safety stock argues that logistics reliability and seasonality should be built into safety-stock logic. By doing that in a live dashboard, the company can hold less dead stock while still protecting service levels during the peak.

Inventory KPI	Before	After AI control	Outcome
Aging stock	Detected late	Aging alert bands	Earlier action
Safety stock	Fixed rules	Seasonality-aware	Less excess
Replenishment	Reactive	Forecast-driven	Fewer rush buys

Warehouse orchestration and peak fulfillment

Business problem: Peak season turns the warehouse into the bottleneck. If inbound, putaway, wave planning, picking, and packing are not coordinated, the team misses ship dates and loses the seasonality advantage. In some cases, the real issue is not lack of product but lack of execution visibility.

Data problem: Warehouse capacity, labor plans, order cutoffs, carrier windows, and backorder priorities often sit in separate tools. That makes it hard to know which orders to prioritize, where labor should move, and which products should be re-slotted closer to the wave picks.

Our solution: Innovacio created a fulfillment dashboard that combines demand forecasts with warehouse execution data. It shows inbound congestion, pick-wave load, cut-off risk, and service-level exposure, then recommends which orders to release, stage, or expedite. The result is a control tower instead of a static WMS screen.

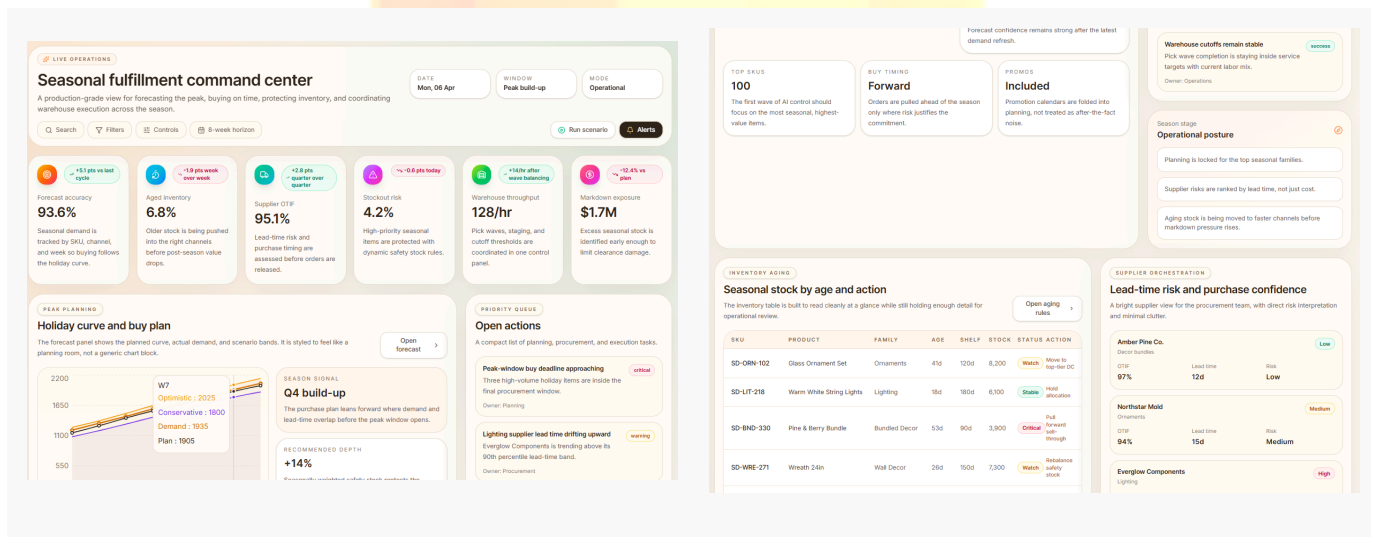
Why it matters: This is where forecasting becomes execution. The literature on forecasting-to-inventory integration is clear that forecasts should not be treated as an end in

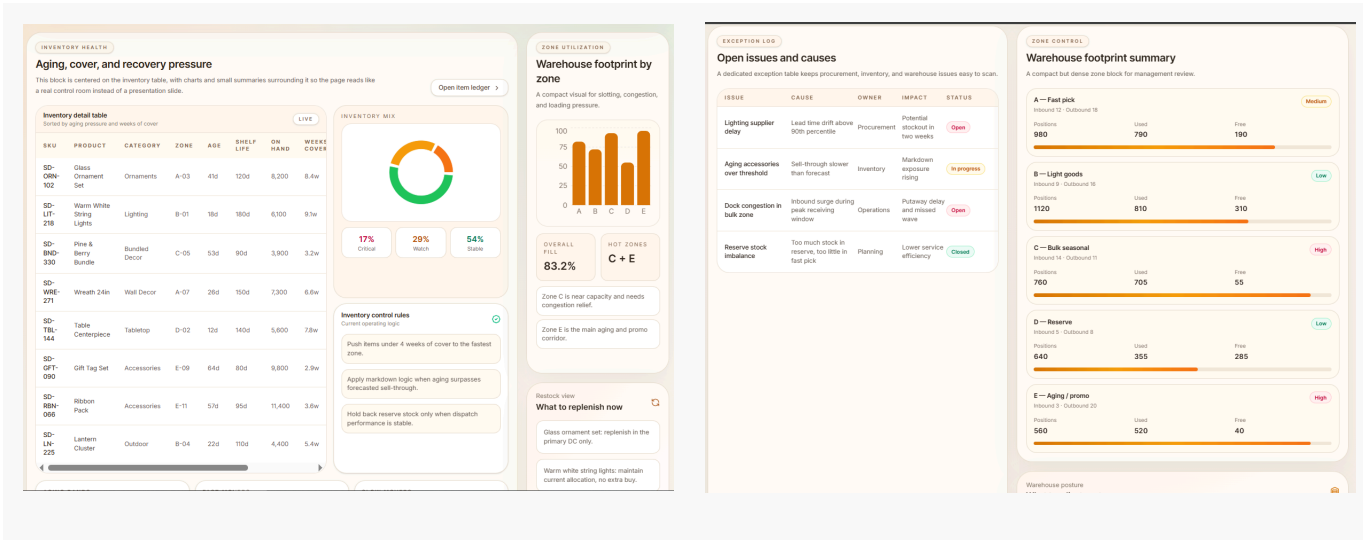


themselves; they need to be converted into replenishment and operating decisions. Innovacio does that conversion inside one operating view.

Fulfillment layer	Before	After AI dashboard	Outcome
Wave planning	Manual and delayed	AI wave priorities	Smoother flow
Cutoff risk	Observed too late	Alerted in real time	Fewer missed ships
Labor focus	Reactive	Workload visibility	Better labor use

Screenshots





Final takeaway

Seasonal décor supply chains are not just “retail with a holiday theme.” They are time-sensitive networks where forecasting, procurement, inventory, and fulfillment all have to move together. Innovacio solves that by building one dashboard that turns seasonality into action: predict the peak, buy on time, protect the inventory, and execute the warehouse plan before the window closes.

Research references used for this case study

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Appendix:

KPI	Why it matters	Evidence source
Forecast accuracy	Shows whether the peak plan is believable	Forecast model output
Sell-through rate	Shows if inventory matches actual demand	Sales and fulfillment data
Aged inventory %	Shows post-season exposure	Inventory aging report
Supplier OTIF	Shows supplier reliability	Supplier scorecard
Stockout rate	Shows service-level protection	Warehouse and order data
Markdown ratio	Shows how much margin is lost after peak	Finance / sales records